

Physiology of Cardiac Cycle – Part 1 – Phases

(Pressure and Volume Changes)

PPOM 2 - Cardiovascular System

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Session Objectives

- 1. Explain pressure/volume changes during each phase of atrial and ventricular systoles and diastoles in a normal cardiac cycle**
- 2. Compare states of heart valves to corresponding phases of cardiac cycle**
- 3. Deconstruct the relationship of electrocardiogram with respect to the events in a cardiac cycle**
- 4. Recognize heart sounds corresponding to a cardiac cycle**
- 5. Map various events of a cardiac cycle in an integrative fashion**

Case study

- A 73-year old man visits ER with complaints of exertional dyspnea, syncope, chest pain. Poorly managed hypercholesterolemia with 11-year history of hypertension.
 - BP: 101/75.
 - Lungs clear. Systolic murmur loudest over the aorta; peaks at mid systole.
 - ECG: LAD, ST normal.
 - CXR: large LV, calcification of aortic valve.
 - Echo: Greatly increased velocity of flow during the systolic portion of the cardiac cycle. Left heart enlarged and LVH.
 - Cardiac enzymes normal.

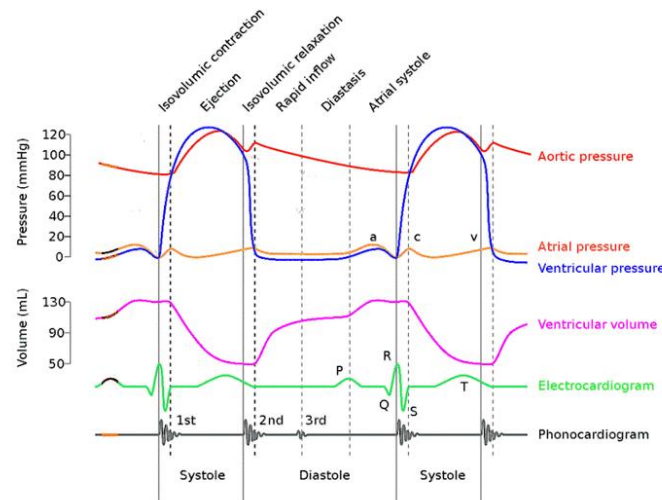
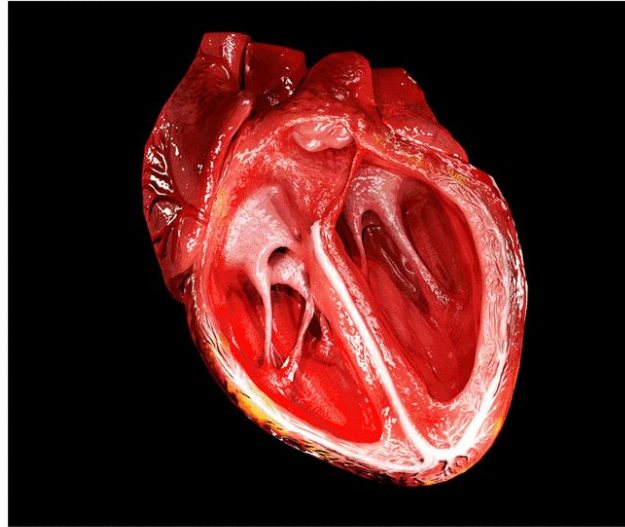
Introduction

- Normal adult Heart Rate: 60-100 bpm
- Cardiac electrical activity → cardiac muscle contraction → blood circulation

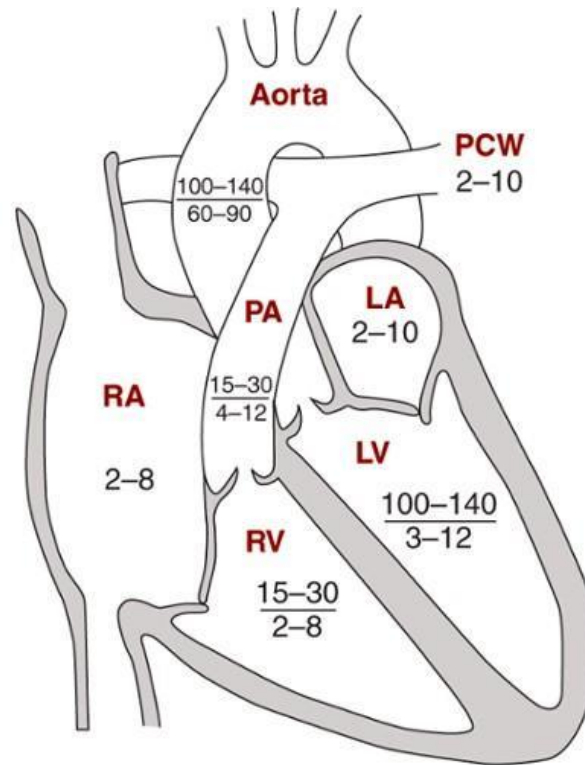
Cardiac cycle: All of the electrical and mechanical events that occur within one heartbeat.

1. Diastole
2. Systole

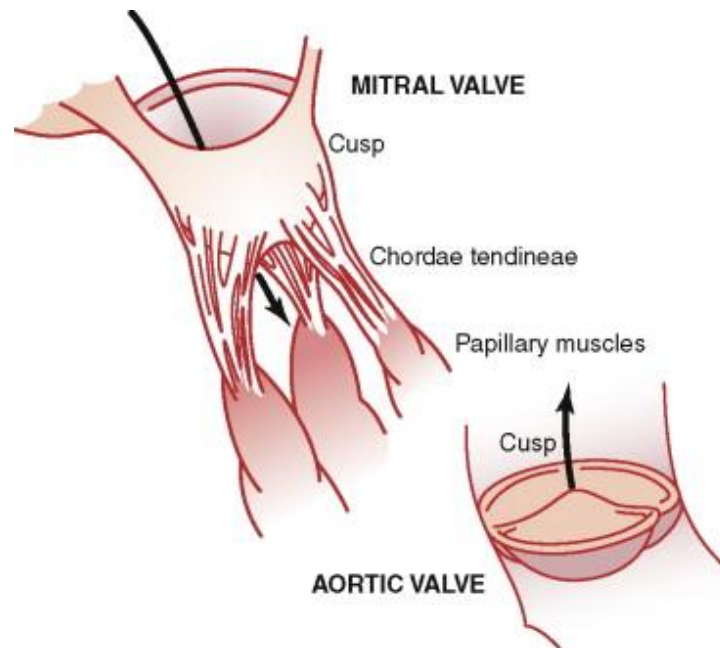
CARDIAC CYCLE



Normal pressures in the cardiac chambers and great vessels



Valves

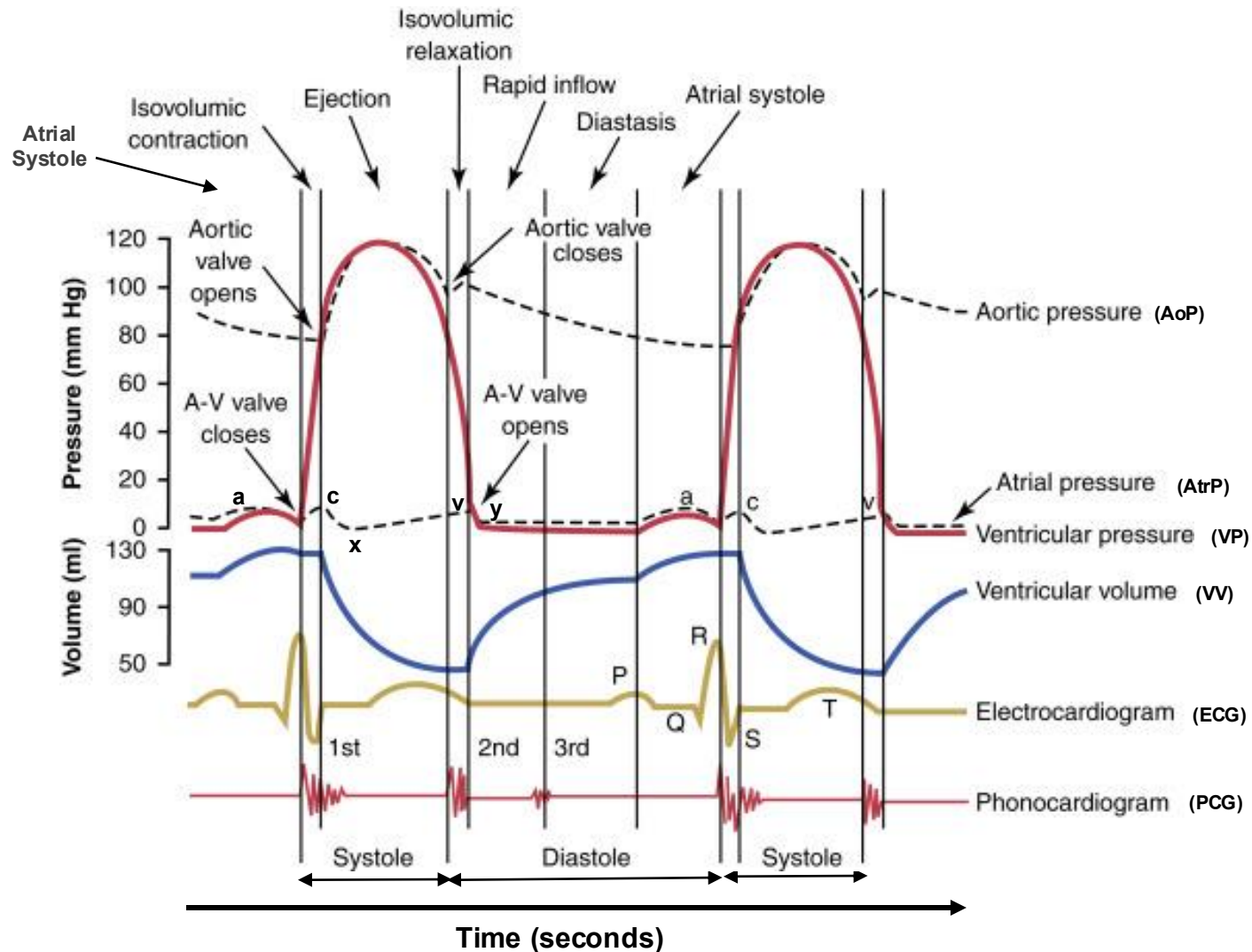


- Beginning of the ventricular systole – the AV valves (e.g., mitral) close
- *Valves are pulled inward toward the ventricles by papillary muscles to prevent their bulging too far backward toward the atria during ventricular contraction*

Background – Cardiac Cycle

- Atrial (LA and RA) contraction **precedes** ventricular (LV and RV) contraction
 - Delay in signal transmission at the AV node $>0.1s$ – helps pump blood into the ventricles before the strong ventricular contraction begins.
- Systole and Diastole – different timings in Atria and Ventricles
- AT REST: **2/3** of the cardiac cycle is spent in **diastole**; 1/3 in systole
- INCREASING ACTIVITY LEVELS:
 - less time in diastole
 - **more time in systole**

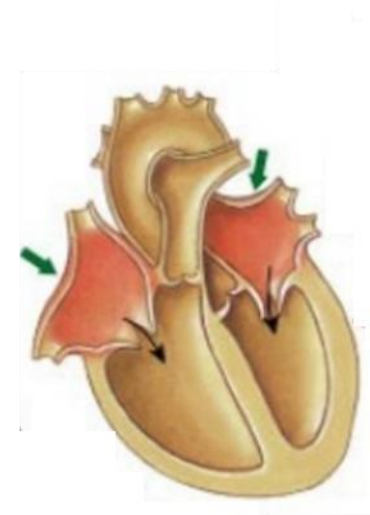
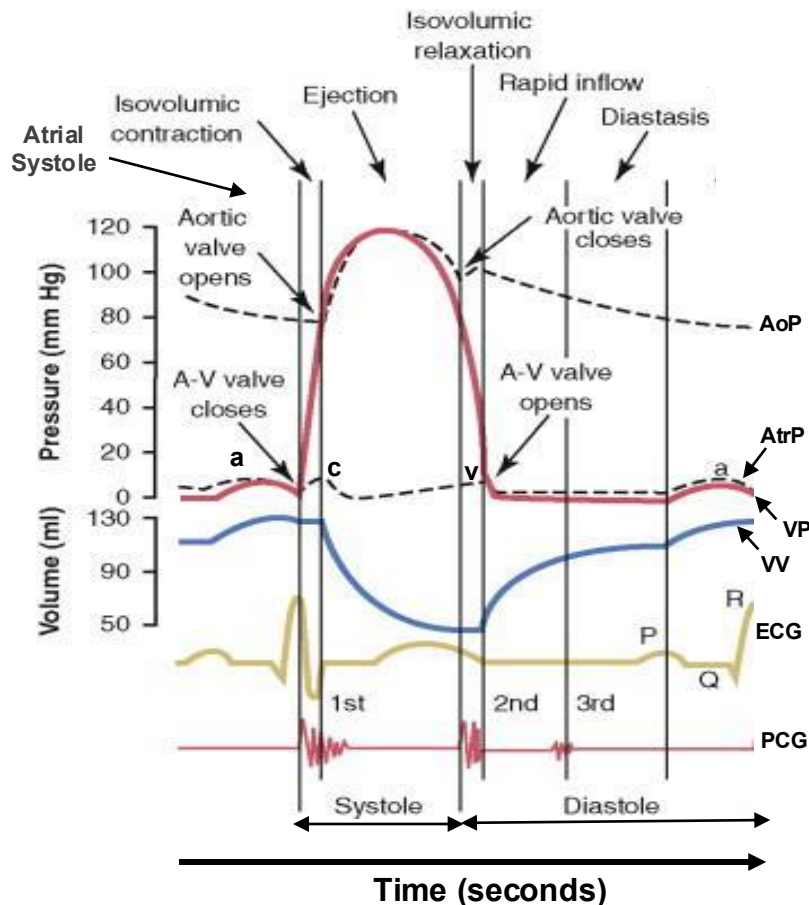
Cardiac Cycle – Wiggers' diagram



Source: Guyton and Hall Textbook of Medical Physiology

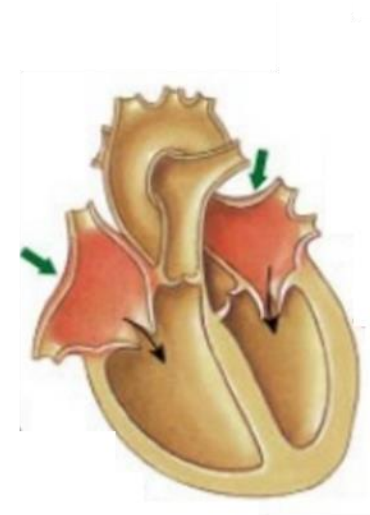
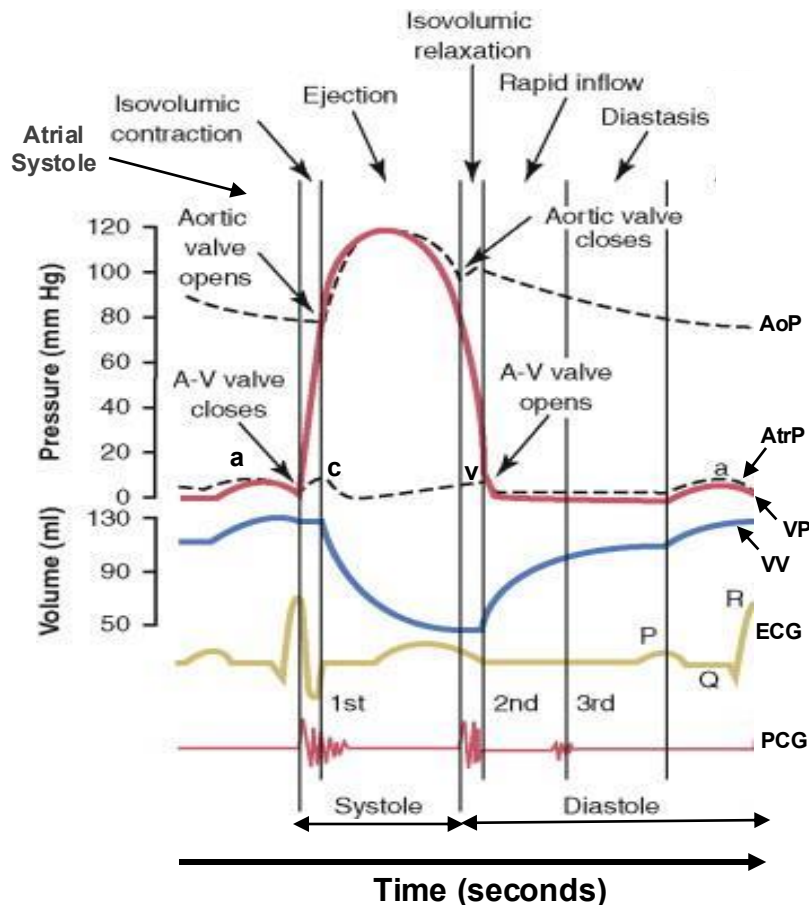
Atrial Systole

- Depolarization of the Sino-Atrial node – not detected
- Atrial depolarization (P-wave) is **required** for and always **precedes atrial contraction**
- *P-wave precedes "a" wave* in the atrial pressure curve by about 0.1s

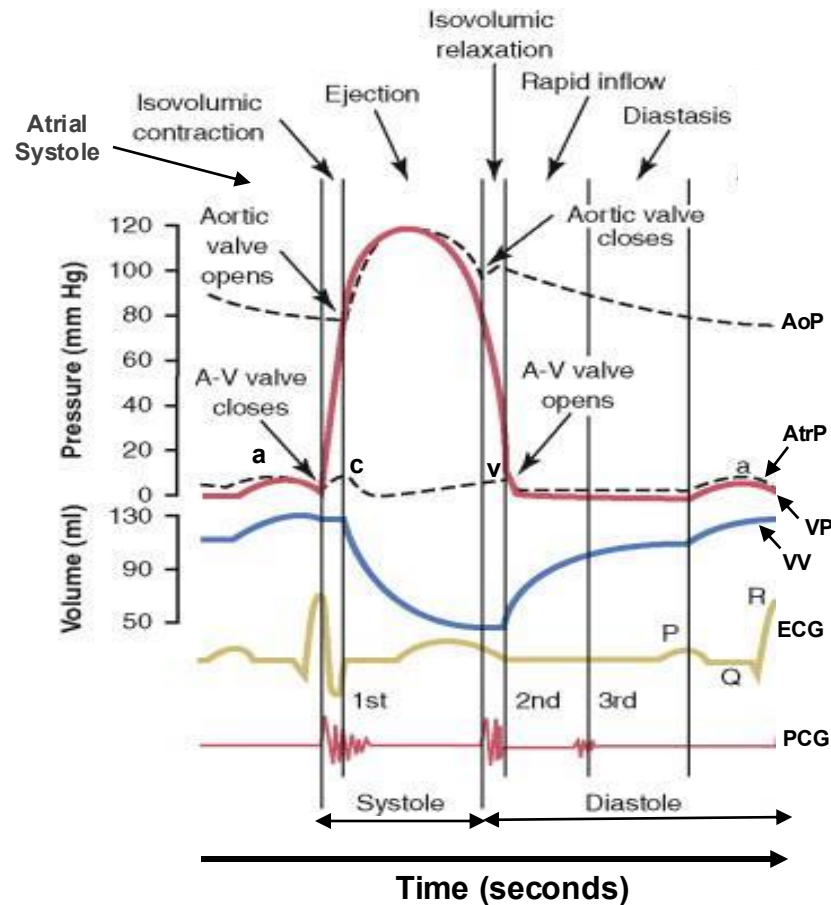


Atrial Systole (Contd...)

- **Atrial contraction** contributes to about **10 ml** (out of about **140 ml**) of LV blood
- **Primer pump**
 - as much as 20%
- **Fibrillation** (irregular/rapid heartbeat)
 - **Atrial** fibrillation

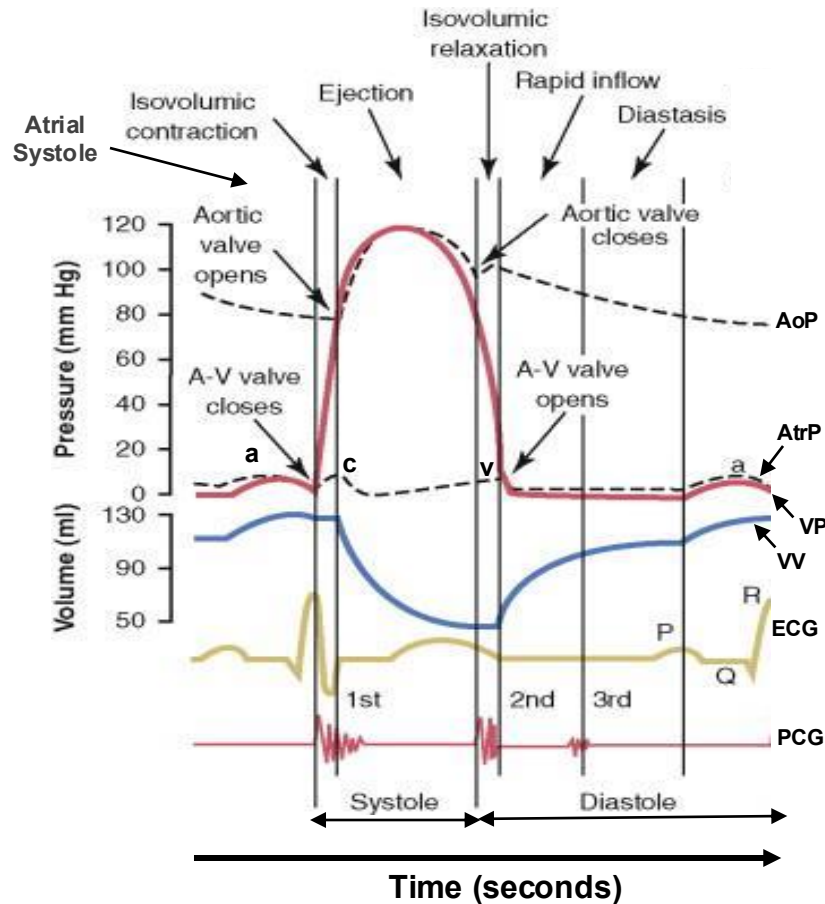


Atrial Diastole



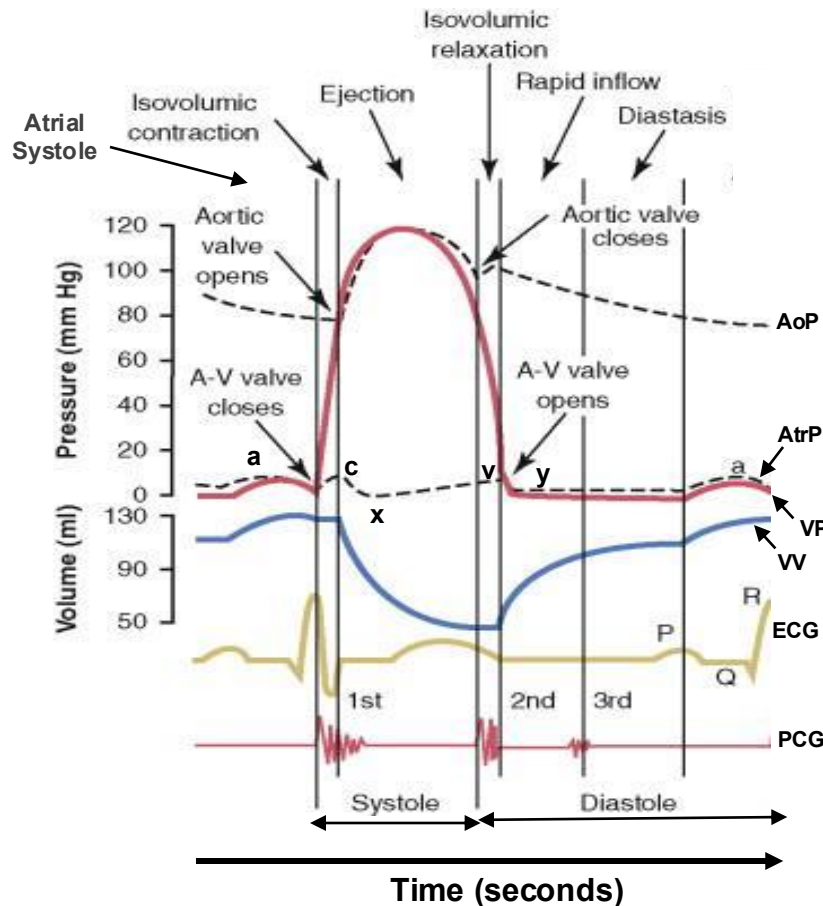
- Atrial diastole
 - during ventricular systole and much of ventricular diastole
- LA Diastolic pressure > RA diastolic pressure (both are low)
- Diastolic blood flow into the atria (from the veins) due to:
 - Pressure difference
 - **Vena cavae/Pulmonary veins > RA/LA**

Atrial Diastole (Contd...)



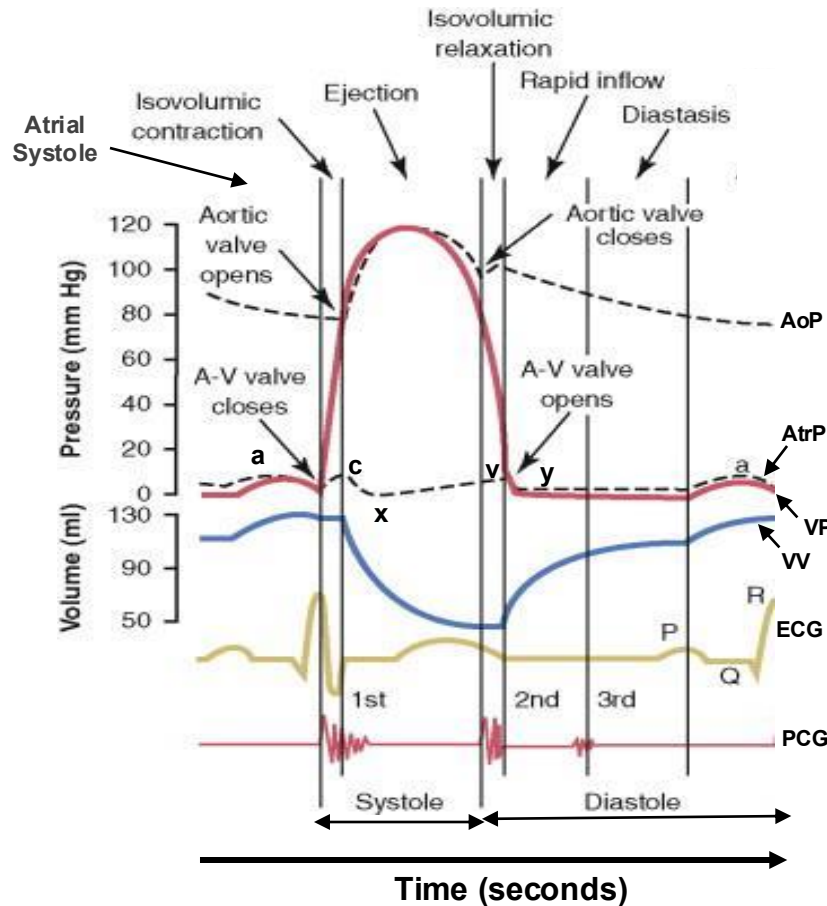
- When the AV valves open, the **small diastolic pressure difference ($A > V$)** permits blood to flow from the atria into the ventricles.
- **Most of the atrial blood enters the ventricle during atrial diastole**

Atrial pressure waves



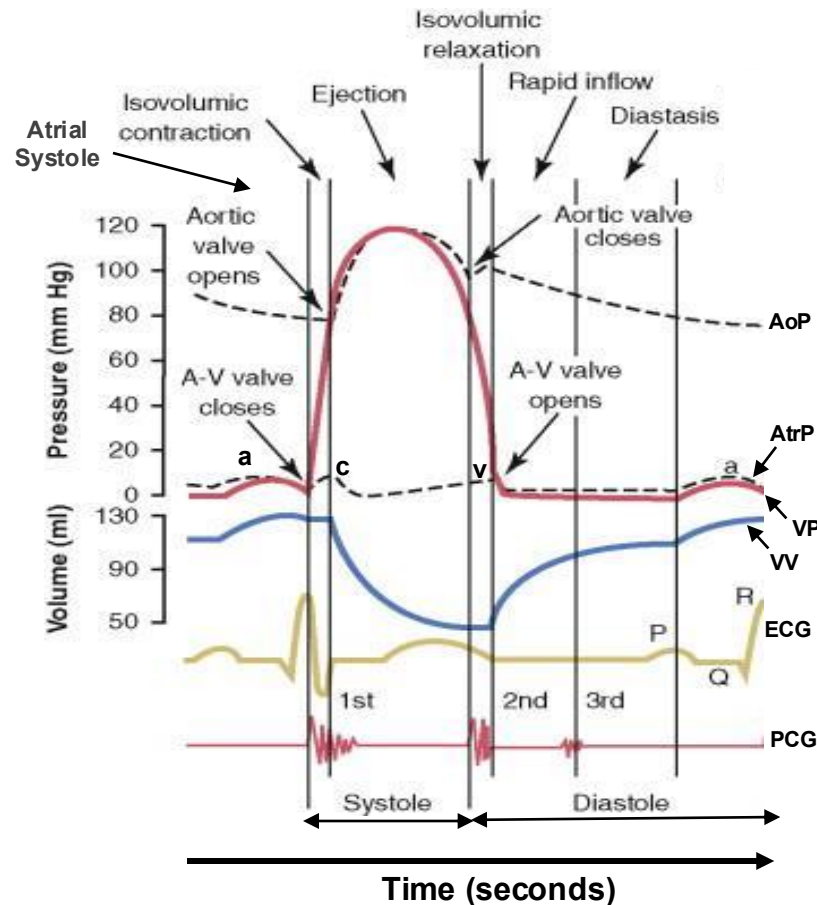
- **"a" wave (a. systole)**
 - small increase in atrial pressure from **atrial contraction**
- **"c" wave (a. diastole)**
 - just after the peak of the QRS complex
 - results from a **slight increase in atrial pressure** due to **atrial compression** during **ventricular contraction**
- **"x" descent**
 - Atrial relaxation

Atrial pressure waves (Contd...)



- **"v" wave (a. diastole)**
 - AV valves closed at the beginning of **ventricular systole**
 - Atrial pressure slowly increases as the **atria fill with incoming blood creating the "v" wave.**
- **"y" descent**
 - Ventricular filling
- AV valves open at beginning of ventricular filling
 - Atrial blood flows into the ventricles
 - Atrial pressure drops

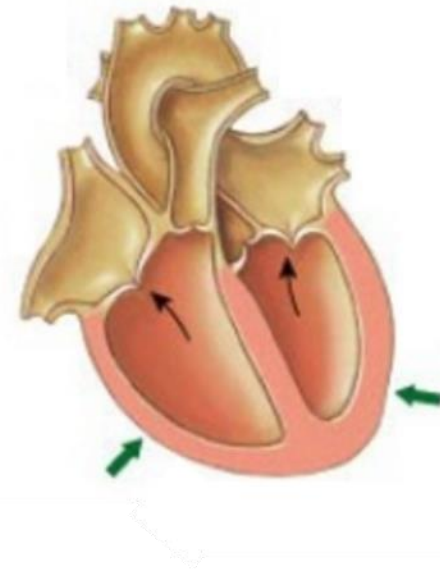
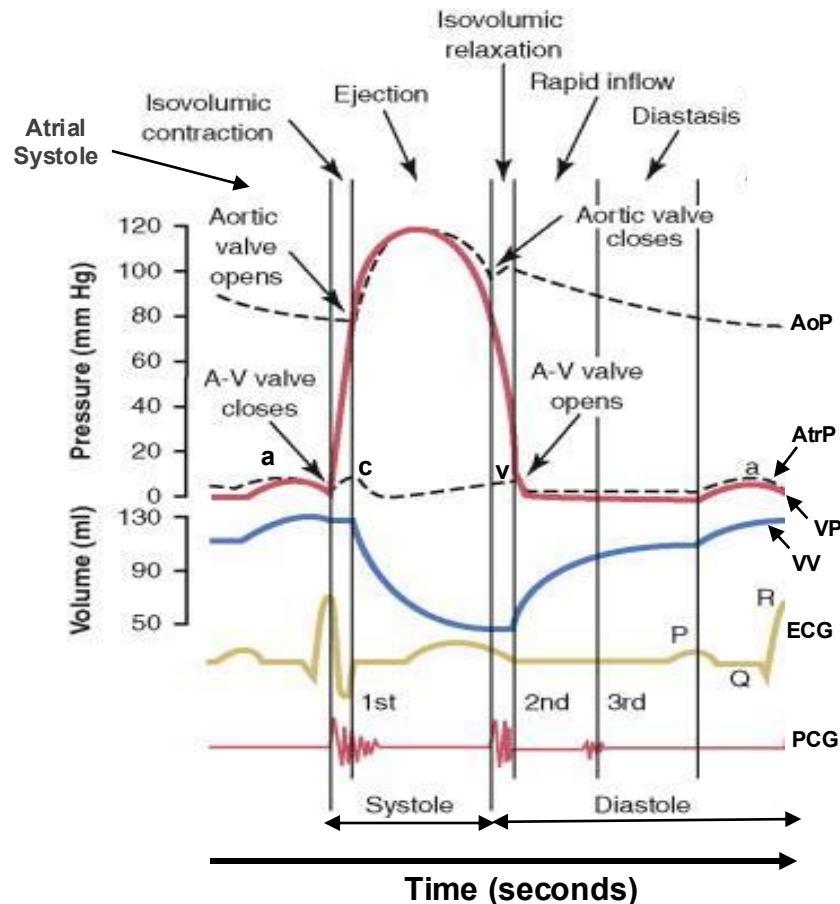
Ventricular systole



- Between start of the QRS complex and (beginning of the) end of the T wave.
- Contraction follows depolarization.
- Ventricular systole occurs in **three phases**
- **End diastolic volume:** Volume of blood in the ventricles at the **start of ventricular contraction (systole)**.
- Ventricular stretch secondary to filling
 - **Frank-Starling law** of the heart (discussed later)

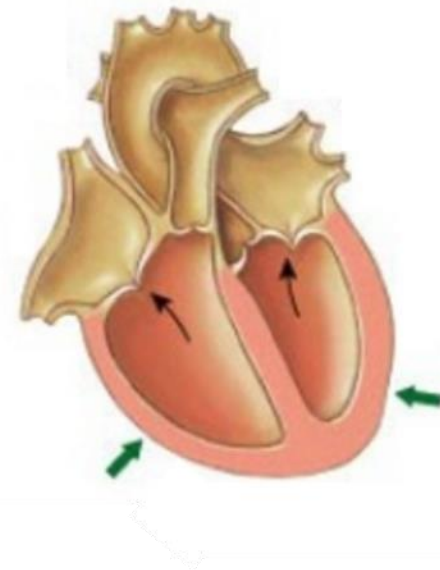
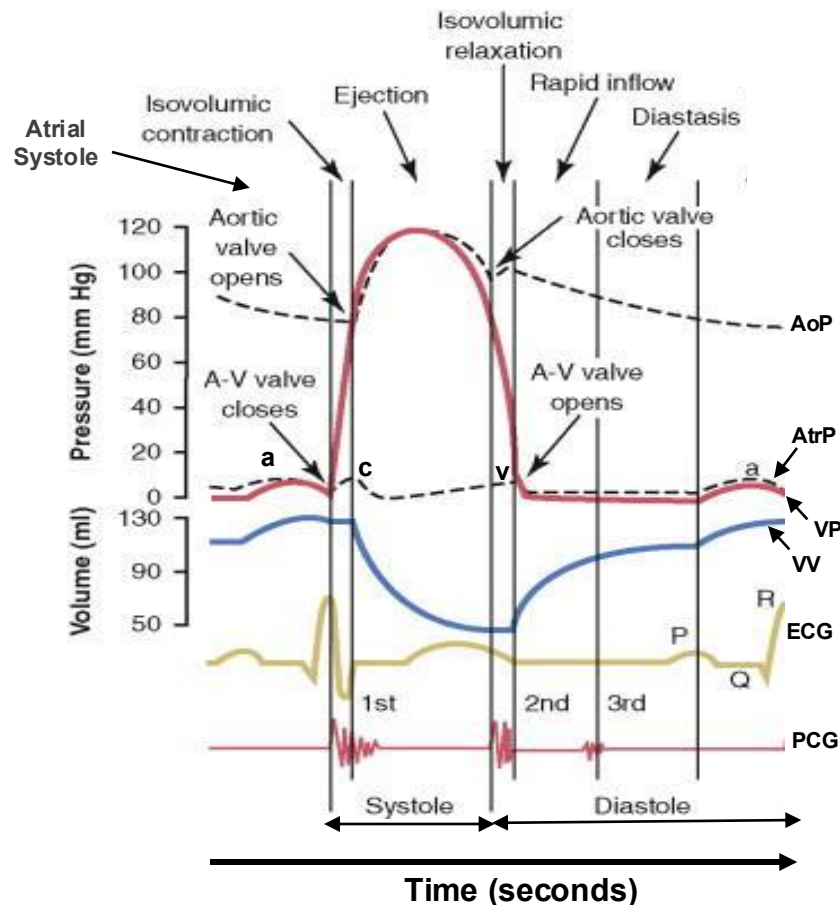
Ventricular systole: Isovolumetric Contraction Phase

- Papillary muscle – initial contraction
- Both valves closed
- Both ventricles are closed, blood-filled chambers
- Since the blood cannot be compressed, the initial effect of ventricular contraction is an **increase in intraventricular pressure without a change in ventricular volume**.



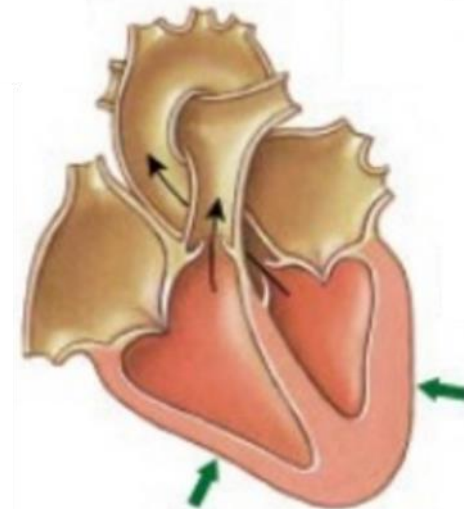
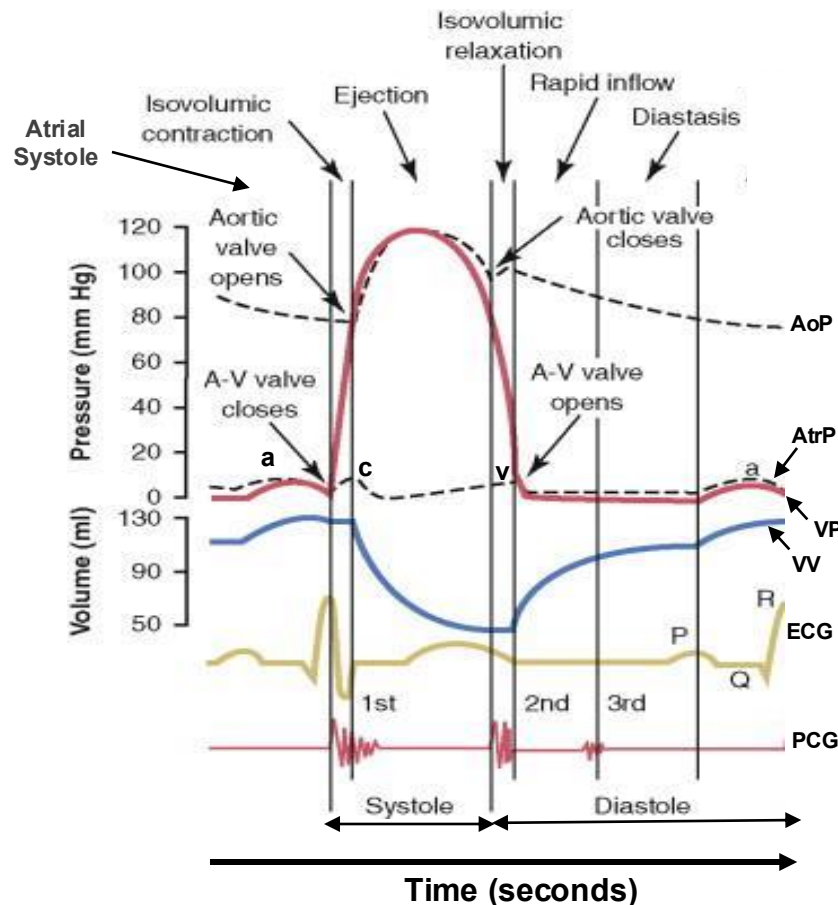
Ventricular systole: Isovolumetric Contraction Phase

- Contraction is isometric
- Both the ventricles develop enough pressure (force) to overcome their respective afterloads (e.g. aortic and PA pressures, discussed later).
- dP/dt (max)



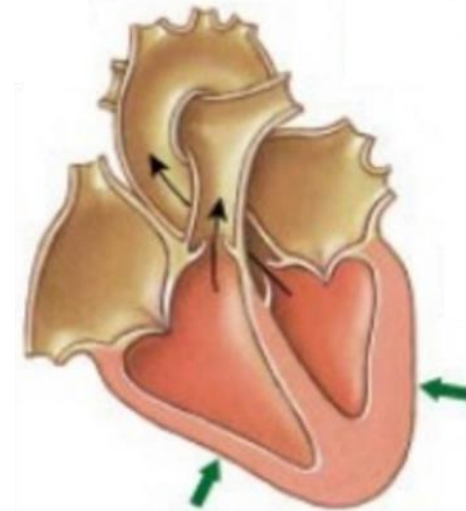
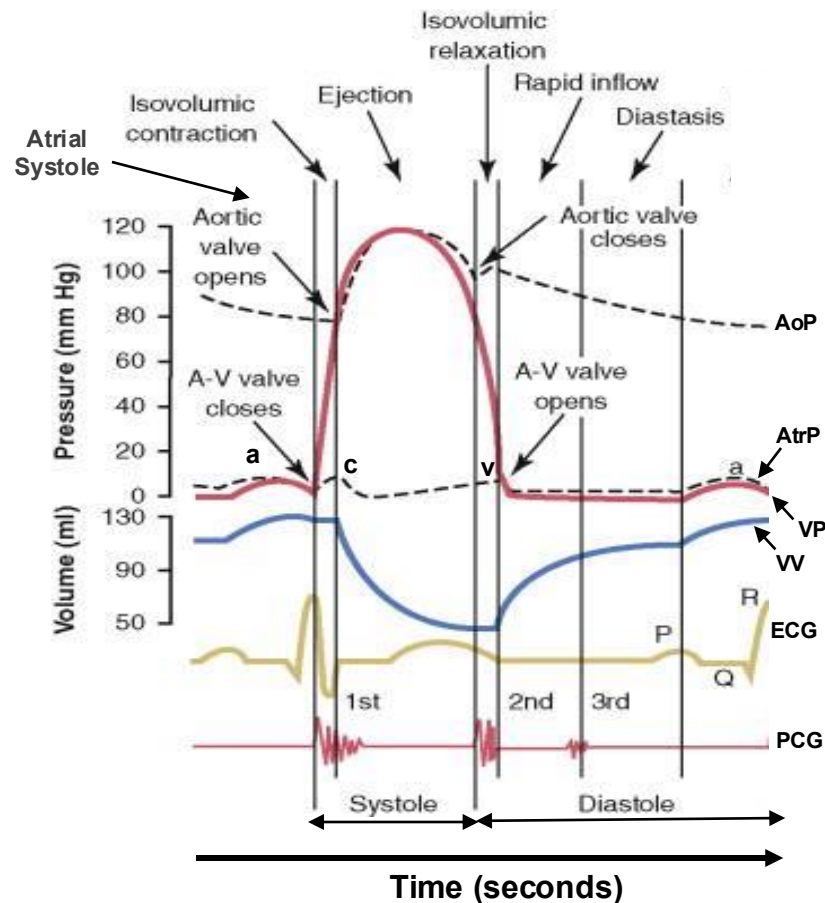
Ventricular systole: Rapid Ejection Phase

- When the **rising ventricular pressure exceeds arterial pressure**.
- The **slightly higher ventricular pressures** force open both semilunar valves and drive blood into the AO or the PA.
- **Ventricular pressure continues to increase** after the AoV/PV have opened because the ventricles are still contracting.



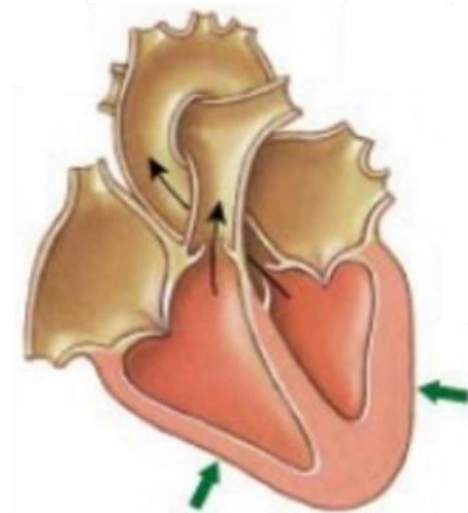
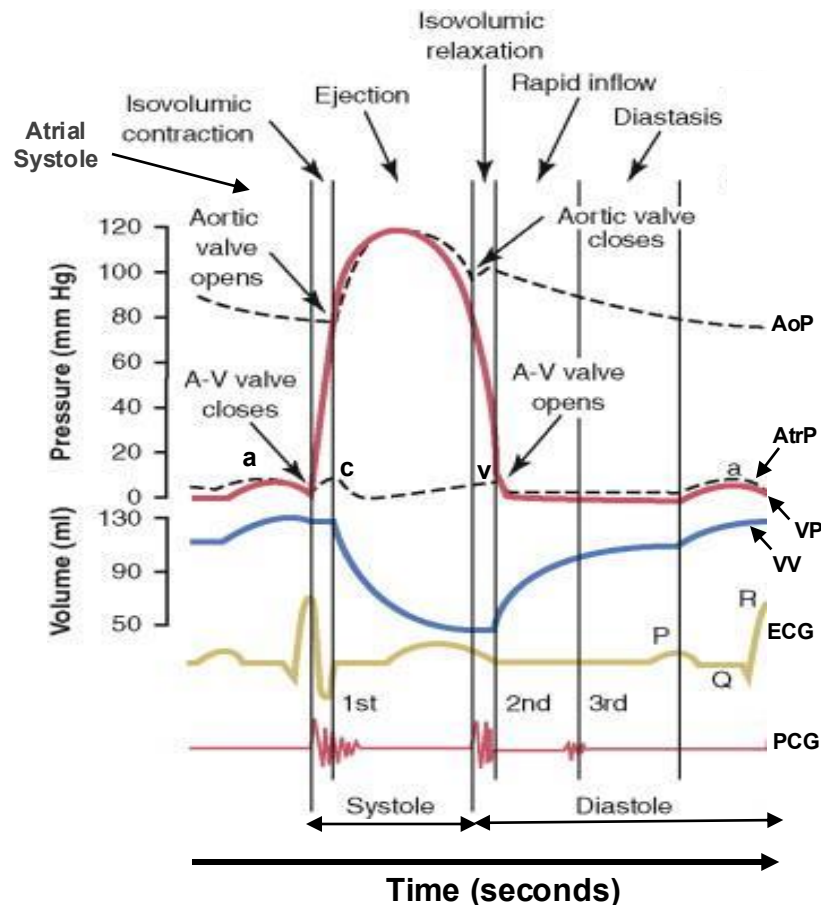
Ventricular systole: Rapid Ejection Phase

- Isotonic contraction
- The aortic (and PA) pressures increase because blood is being forced into them.
- LVP > BP



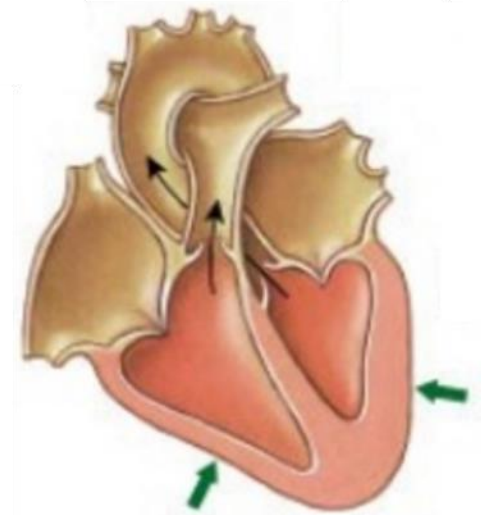
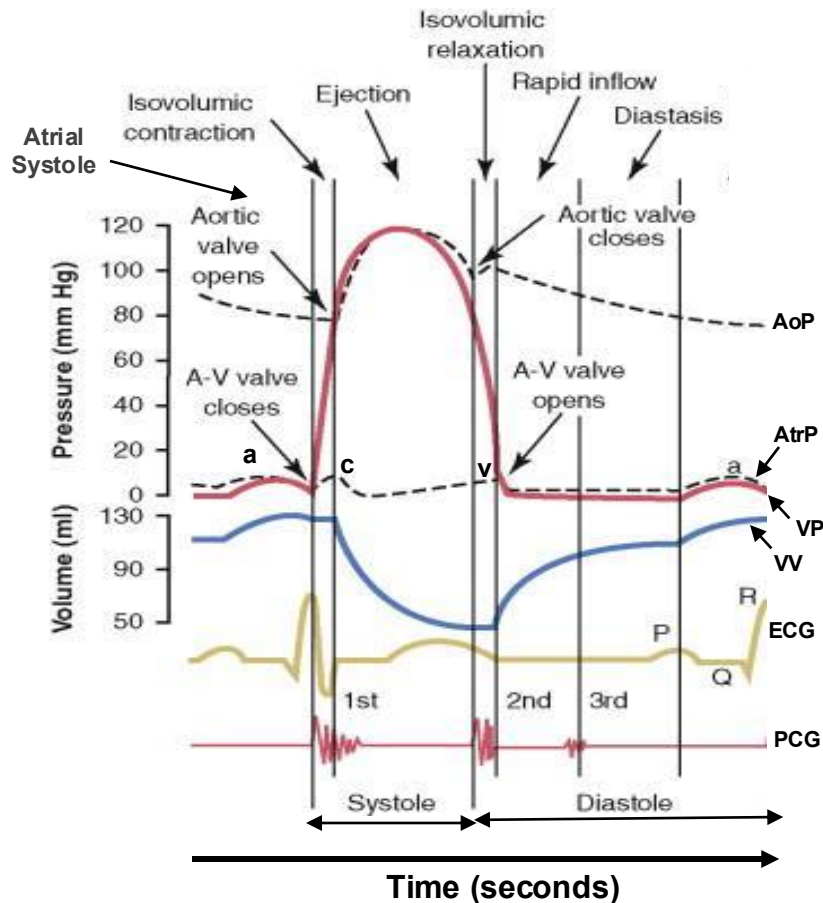
Ventricular systole: Reduced Ejection Phase

- **LVP falls** slightly below AO (or PA) pressure
- Blood is still ejected from the LV.
- The **inertia imparted to the blood by ventricular contraction**, rather than the pressure difference is now the **driving force for blood ejection**.



Ventricular systole: Reduced Ejection Phase (Contd...)

- This phase stops when ventricular pressure falls below the levels needed to keep the AoV/PV open.
- Coincides with the T wave of the ECG.



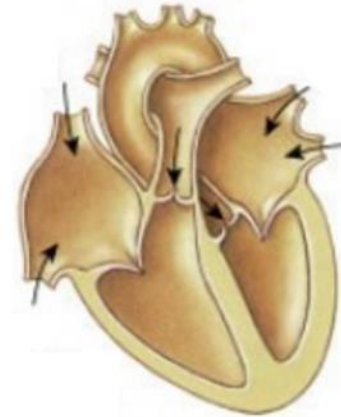
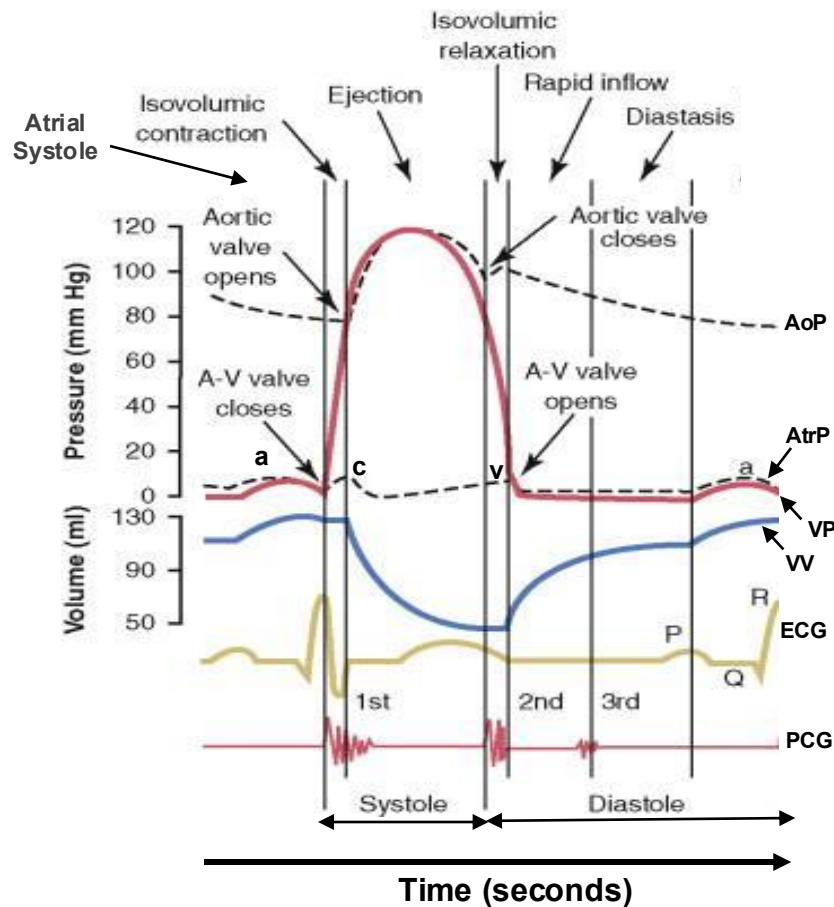
Sample question 1

The rapid ventricular ejection phase changes to the reduced ejection phase when?

- A. Aortic pressure reaches its minimum diastolic value.
- B. Left semilunar valve closes
- C. Left AV valve opens
- D. Pressure in the left ventricle falls below the pressure in the aorta.
- E. Atrial pressure becomes greater than ventricular pressure.

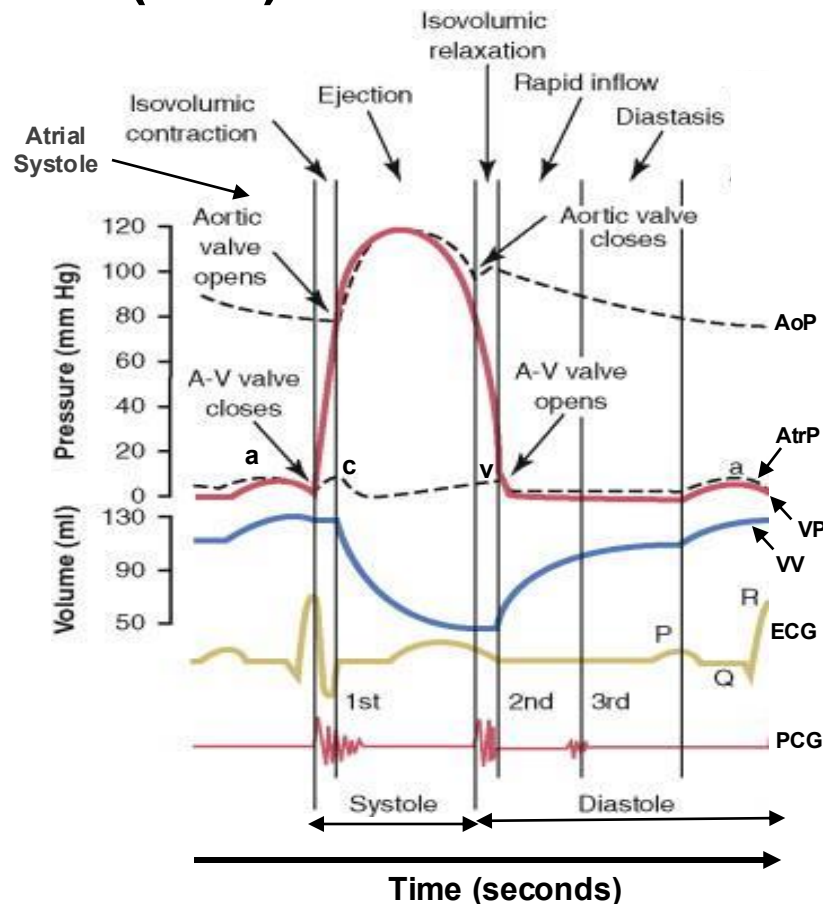
Ventricular diastole: Isovolumetric Relaxation Phase

- Ventricular diastole (VD) – three phases:
- VD starts with the **closure of the semilunar valves** (AV valve also closed)
- Isovolumetric relaxation**



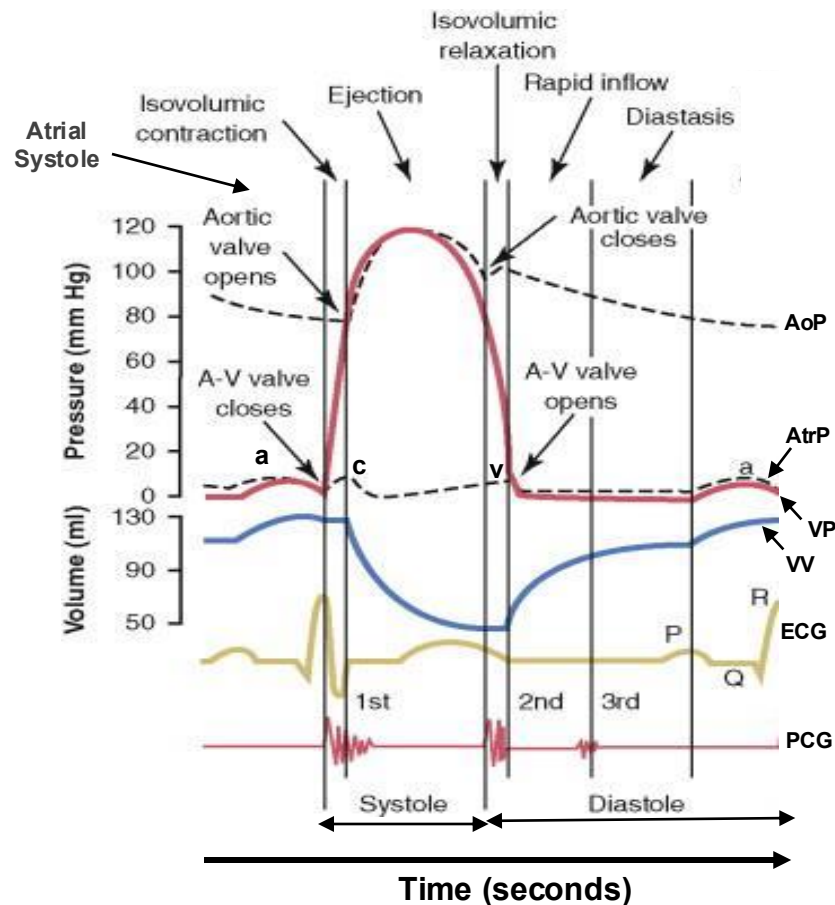
Ventricular diastole: Isovolumetric Relaxation Phase

- Fall in ventricular pressure follows T wave repolarization.
- LV does not completely empty during ejection phase.
- The **volume** of blood remaining in the ventricles at the **end of systole (or start of diastole)** is called end systolic volume (ESV).
- **dP/dt (min)**



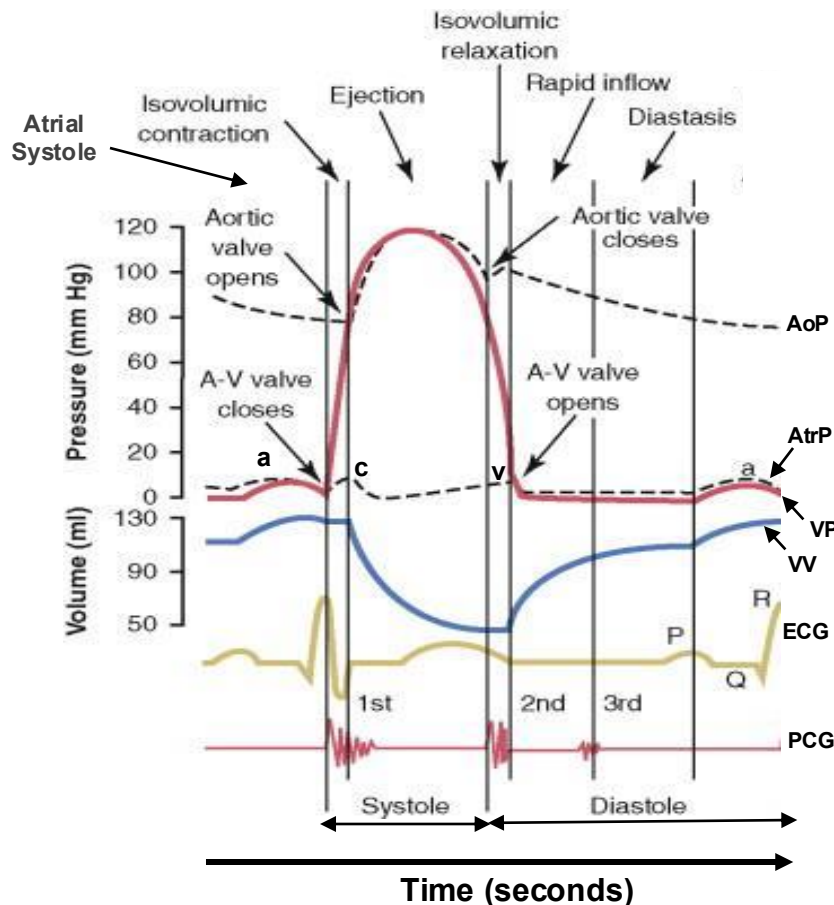
Ventricular diastole: Rapid filling phase

- VD pressure falls below atrial pressure
- The pressure difference causes the AV valves to open

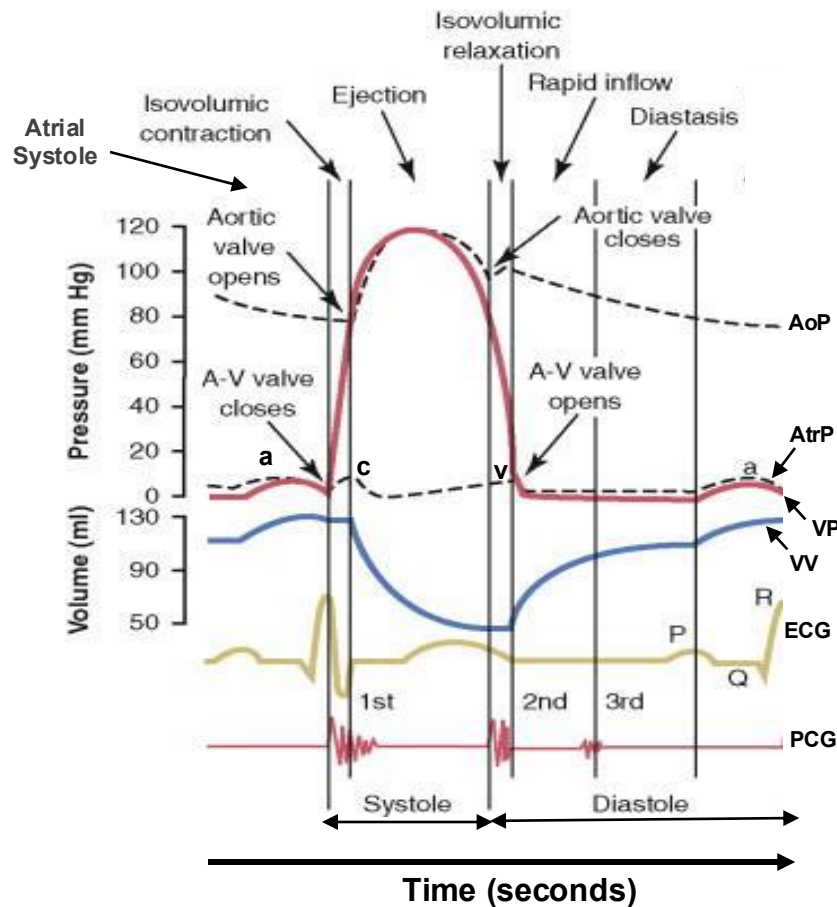


Ventricular diastole: Reduced filling phase

- Blood flow: Veins → relaxed atria → relaxed ventricles
- VD Pressure << Atrial pressure, VD transitions into a slower **reduced filling phase (or diastasis)**.
- The **Diastasis** ends with the **P wave** of the next cycle.
- VD ends with the appearance of R-wave

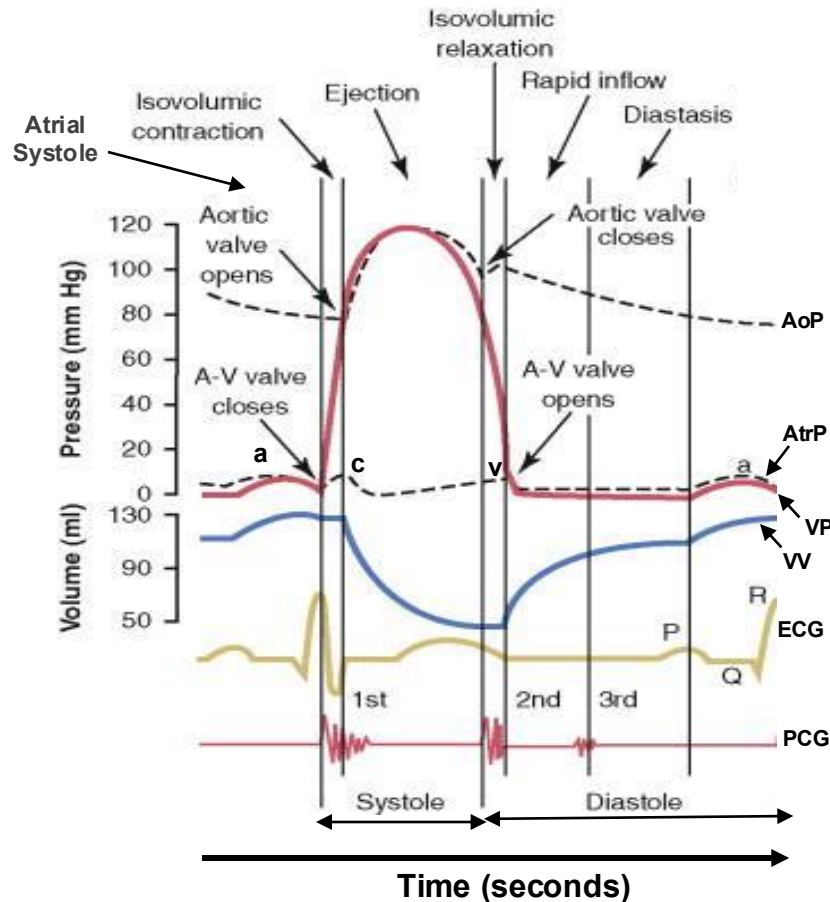


Heart Sounds



- Four heart sounds
- In normal adults, only S1 and S2 heard
- **S1** is associated with **closure of the AV valves**
- **S2** is associated with **closure of the semilunar valves**

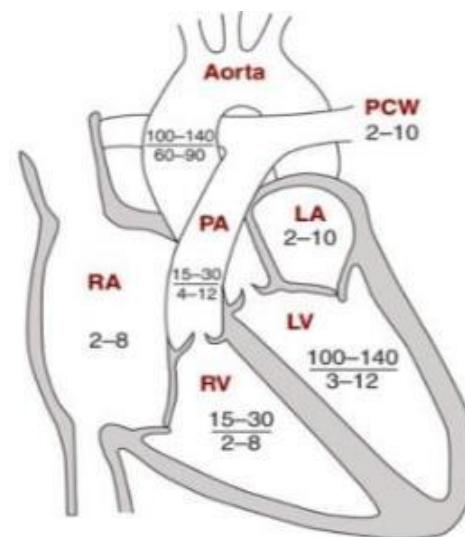
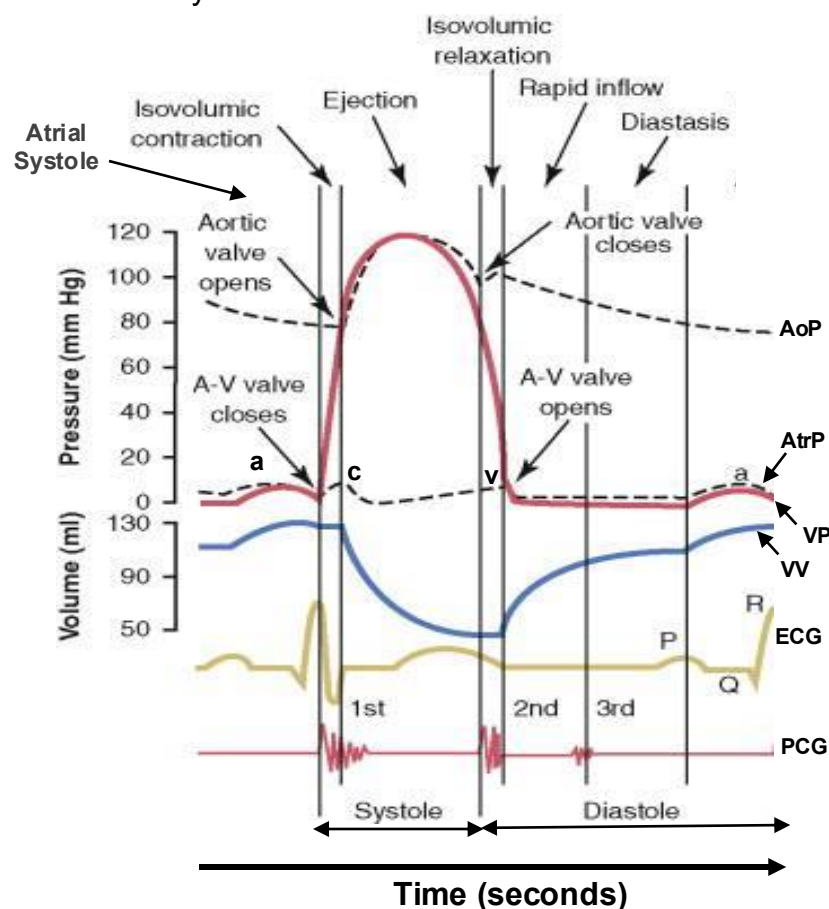
Heart Sounds (Contd...)



- **S3** gallop is associated with the **rapid filling phase** of ventricular diastole. It is frequently heard in normal children.
- **S4** gallop is associated with **atrial systole (end-diastolic)** against a hypertrophied ventricle

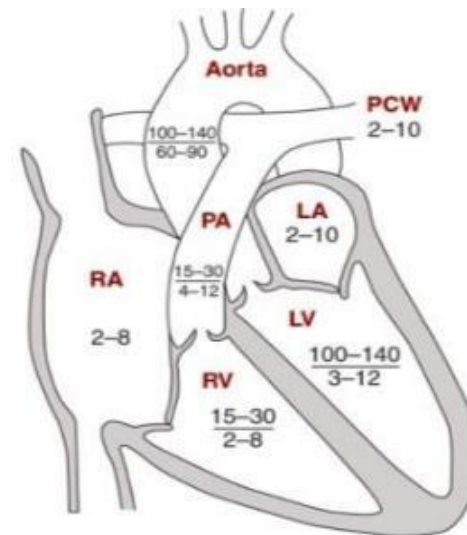
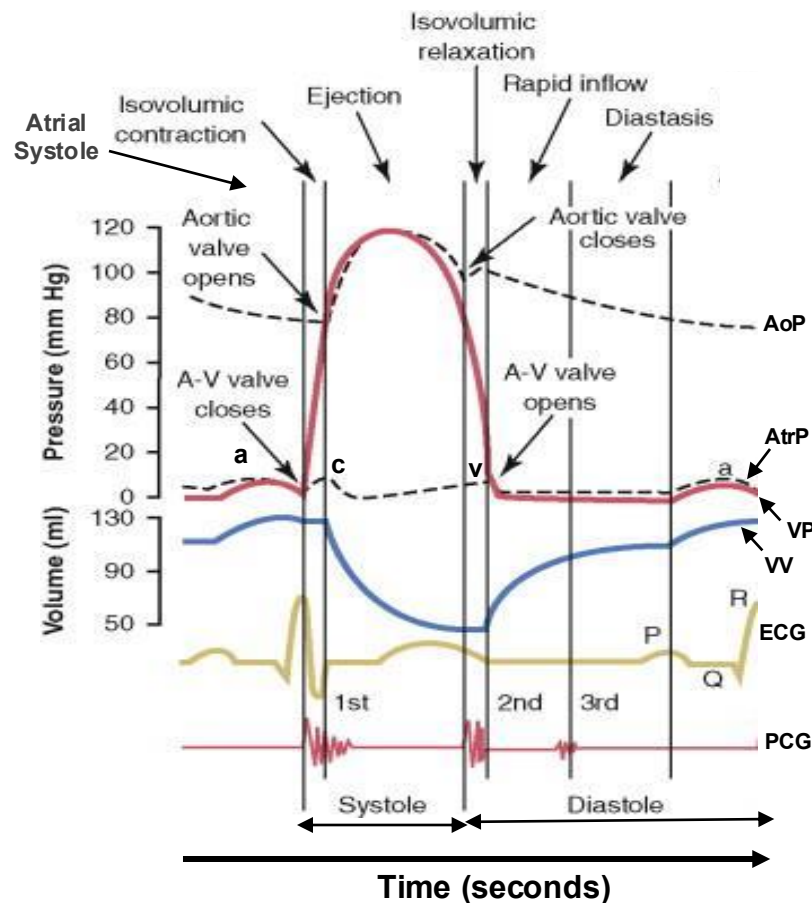
Aortic pressure changes and Pulse pressure

- Max syst. Ao Pressure ~ 120 mm Hg at rest.
- **Rapid** ejection phase – LVP > Syst. Ao. Pressure
- **Reduced** ejection phase – Systolic Ao. Pressure > LVP
 - Ao pressure decreases as LV pressure decreases **until closure of the semilunar valve.**
 - The **dicrotic notch** or **incisura** is caused by a slight back flow of aortic blood that fills the cusps of the SV as they close.

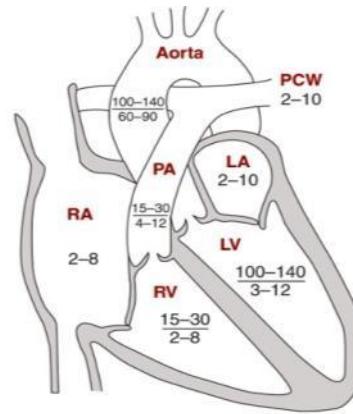
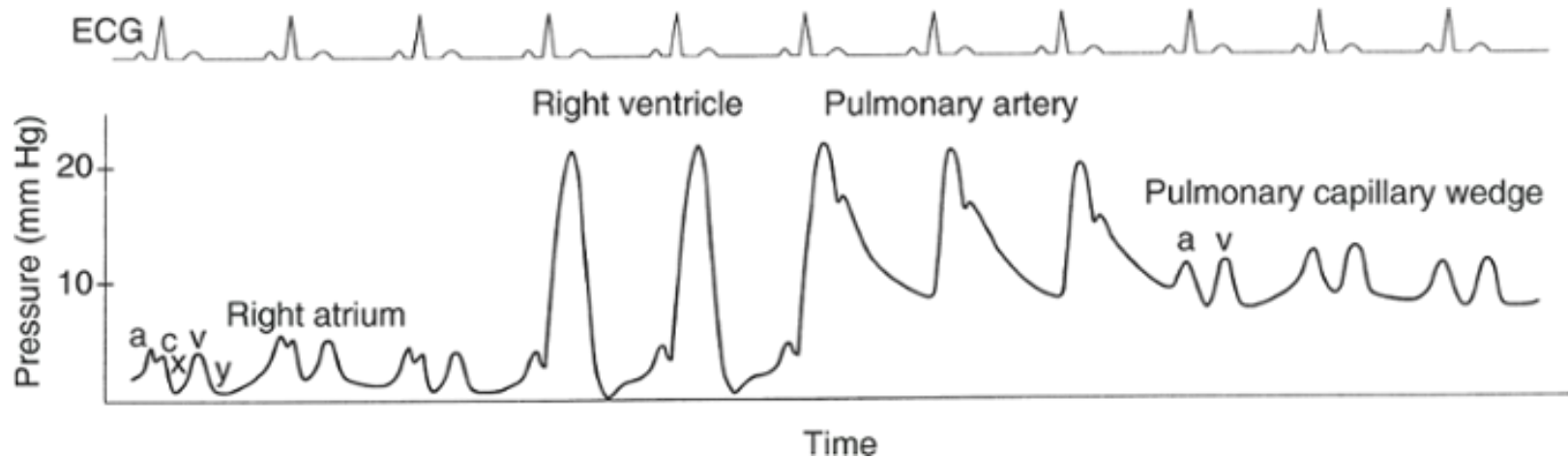


Aortic pressure changes and Pulse pressure (contd...)

- **Ventricular diastole**
 - **Ao Pressure gradually decreases**, and **reaches** its lowest diastolic value, (**~ 80 mmHg**), just before opening of the SV in the next cycle.
- **Blood flows** through the **arteries in spurts** because of the variation in Ao pressure. **Flow in capillaries and veins is steady**.



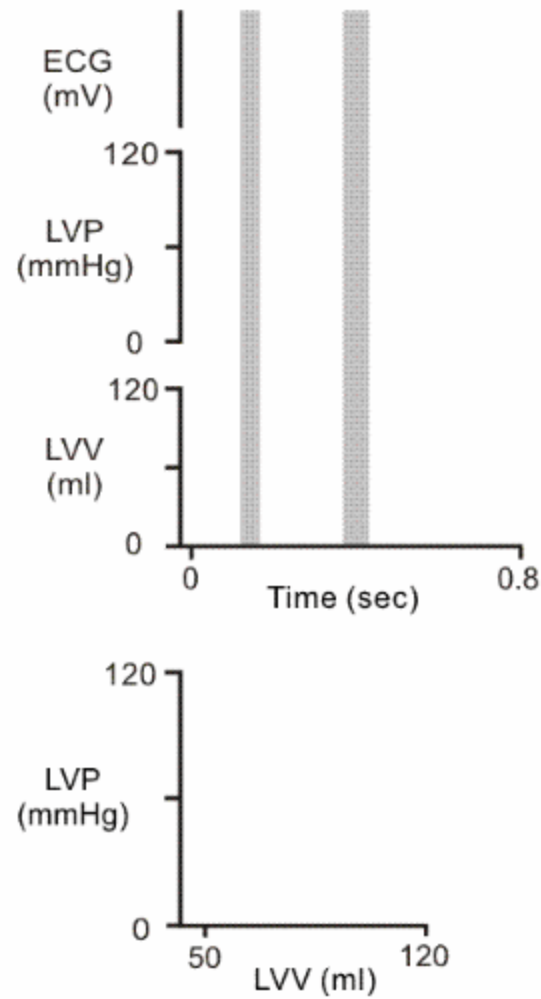
Right-sided pressures



Summary

Phase of Cardiac Cycle*	Major Events	Electrocardiogram	Valves	Heart Sounds
Atrial Systole (A)	Atria contract Final phase of ventricular filling	P wave PR interval	—	Fourth heart sound
Isovolumetric Ventricular Contraction (B)	Ventricles contract Ventricular pressure increases Ventricular volume is constant (all valves are closed)	QRS complex	Mitral valve closes	First heart sound
Rapid Ventricular Ejection (C)	Ventricles contract Ventricular pressure increases and reaches maximum Ventricles eject blood into arteries Ventricular volume decreases Aortic pressure increases and reaches maximum	ST segment	Aortic valve opens	—
Reduced Ventricular Ejection (D)	Ventricles eject blood into arteries (slower rate) Ventricular volume reaches minimum Aortic pressure starts to fall as blood runs off into arteries	T wave	—	—
Isovolumetric Ventricular Relaxation (E)	Ventricles relaxed Ventricular pressure decreases Ventricular volume is constant	—	Aortic valve closes	Second heart sound
Rapid Ventricular Filling (F)	Ventricles relaxed Ventricles fill passively with blood from atria Ventricular volume increases Ventricular pressure low and constant	—	Mitral valve opens	Third heart sound
Reduced Ventricular Filling, or Diastasis (G)	Ventricles relaxed Final phase of ventricular filling	—	—	—

Source: Physiology by Linda S Costanzo



Sample question 2

During ventricular ejection phase, which of the following statements correctly describes the valve movement?

- A. The aortic valve is closed and the left AV valve is open.
- B. The aortic valve is open and the left AV valve is open.
- C. The aortic valve is closed and the left AV valve is closed.
- D. The right and left AV valves are open.
- E. The aortic valve is open and the left AV valve is closed.

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More sample questions for practicing

1. <https://drive.google.com/file/d/1XBNAGbhbRnIVyGtPFaRnR-ygLB1RUAsZ/view?usp=sharing>
2. <https://exchange.scholarrx.com/brick/cardiac-cycle>

References

- **Guyton and Hall, Textbook of Medical Physiology, 14th and 15th Eds.**
 - Chapter 9 - Cardiac Muscle; The Heart as a Pump and Function of the Heart Valves
 - Online - Copy & paste should work; direct-click may not work sometimes
 - <https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C20170004883>
 - <https://www-clinicalkey-com.nyit.idm.oclc.org/#!/browse/book/3-s2.0-C20220008424>
- **ScholarRx Brick(s) and link(s):**
 - Cardiac Cycle:
 - <https://exchange.scholarrx.com/brick/cardiac-cycle>



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